

Lucas Zheng

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EDUCATION

Brown University

Providence, RI

B.S. in Computer Science & B.A. in Applied Mathematics; GPA: 4.00/4.00

Aug. 2023 – May 2027

- Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Sequential Decision Making

EXPERIENCE

Research Assistant

Sep. 2025 – Present

Intelligent Robot Lab, Brown University

Providence, RI

- Building a hierarchical reinforcement learning system under Prof. George Konidaris with concurrent training of option-selection agents and option-policy agents (DQN, PPO), evaluating both learned and fixed options on Montezuma's Revenge and MiniGrid
- Researching exploration strategies including count-based intrinsic motivation methods to address sparse-reward, long-horizon challenges in Montezuma's Revenge where standard exploration fails

Software Engineering Intern

Jul. 2025 – Aug. 2025

Supio

Seattle, WA

- Reduced deposition transcript turnaround time from 10 days to minutes by engineering an end-to-end pipeline with Whisper, pyannote speaker diarization, and an OpenAI API correction workflow
- Integrated transcripts into hybrid RAG search system combining pgvector semantic similarity with BM25 ranking algorithms, enabling lawyers to query hours of audio content with sub-second response times

Research Intern

Jul. 2024 – Aug. 2024

Machine Creativity Lab, Hong Kong University of Science and Technology

Hong Kong

- Advanced lab's generative music AI's capabilities by training an audio emotion classifier with PyTorch and boosting classification accuracy by 6% through fine-tuning
- Engineered feature extraction pipeline processing 10,000+ audio samples with automated labeling algorithms and statistical validation, leading to integration into lab's production music AI model

Natural Language Processing Intern

Jun. 2023 – Aug. 2023

Otter.ai

Mountain View, CA

- Trained and fine-tuned DeBERTa models for customer classification tasks on 5000+ customer transcripts using PyTorch and scikit-learn, providing key data on customer segmentation that contributed to the strategic decision to refocus the product roadmap on general meeting over sales use cases

PROJECTS

AI Restaurant Reservation Voice Agent | *Python, FastAPI, Gemini API, Twilio*

2025 – Present

- Built a voice AI agent using Gemini Flash and Twilio that autonomously calls restaurants, negotiates availability, and books reservations through real-time natural conversation
- Engineered a cascade orchestrator with async state management and Redis for multi-restaurant failover, and bidirectional WebSocket audio streaming with real-time sample rate conversion between Twilio and Gemini API

Physics-Informed Neural Network for Seismic Localization | *Python, PyTorch, Gymnasium*

2025

- Developed a Physics-Informed Neural Network (PINN) with Fourier Feature Encoding to solve the Eikonal equation for seismic localization and a PPO Reinforcement Learning agent to optimize adaptive collocation sampling by formulating the training loop as a Markov Decision Process

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Tools & Frameworks: PyTorch, pandas, Gymnasium, scikit-learn, PostgreSQL, MCP, Git, REST APIs

MISCELLANEOUS

Hackathons: 2nd Place winner at [2026 Precision Neuroscience BCI Hackathon](#)

Languages: Mandarin (fluent), Arabic (working proficiency – U.S. State Department NSLI-Y Scholar)

Interests: Squash, Golf, Poker, Chess